

FUNCTIONAL DISORDERS

Impact of Bile Acid Diarrhea in Patients With Diarrhea-Predominant Irritable Bowel Syndrome on Symptoms and Quality of Life

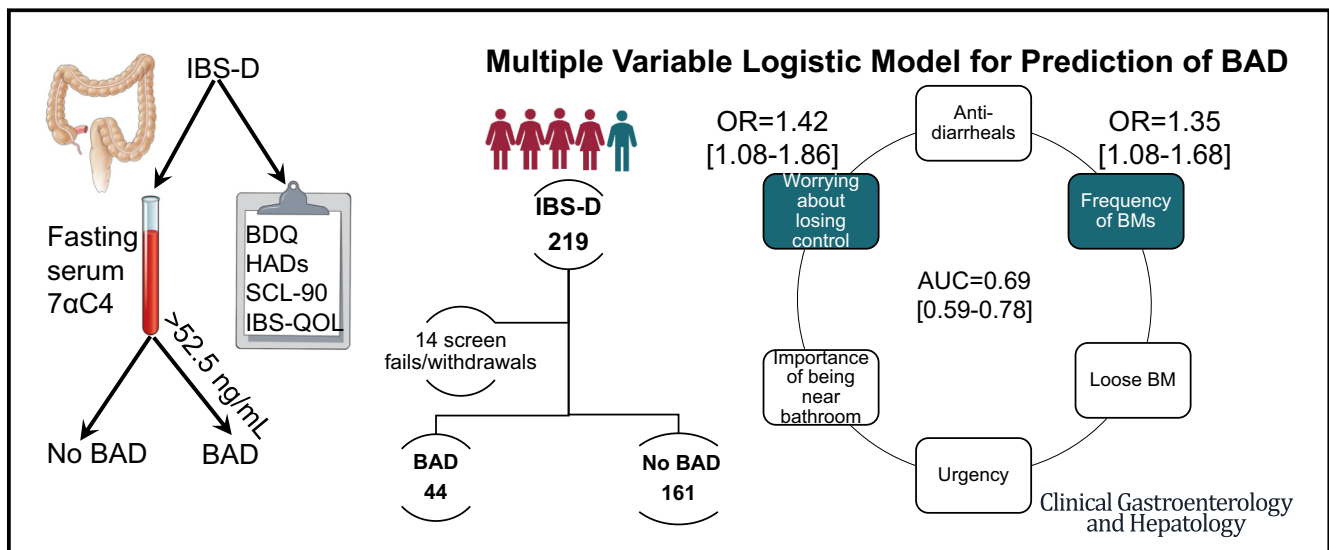


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This article has an accompanying continuing medical education activity, also eligible for MOC credit on page e1533. Upon completion of reading this article, successful learners will be able to identify clinical features that may differentiate patients with bile acid diarrhea among those presenting with symptoms suggestive of irritable bowel syndrome with diarrhea.



BACKGROUND & AIMS:

Bile acid diarrhea (BAD) affects approximately a quarter of patients with irritable bowel syndrome with diarrhea (IBS-D). We aimed to compare the demographics, bowel and somatic symptoms, and quality of life of patients with IBS-D, with or without BAD.

METHODS:

On one occasion, patients with IBS-D (positive for Rome III criteria) completed the following questionnaires: bowel disease questionnaire, Hospital Anxiety and Depression inventory, general quality of life (Symptom Checklist-90), and IBS-specific quality of life. A fasting serum C4 level higher than 52.5 ng/mL was used as a biomarker for BAD. Statistical analysis included a multiple variable logistic model to identify strong predictors of BAD in IBS-D.

RESULTS:

Among 219 patients (79% female) with IBS-D, 44 had BAD; the BAD group was significantly older and had a higher body mass index than the patients without BAD. Patients with BAD had more severe bowel dysfunction and impact on IBS-specific quality of life (need of toilet proximity) compared with patients with IBS-D without BAD. Patients with BAD were more likely than other IBS-D groups to receive antidiarrheals, bile acid binders, and antacid secretory agents. The severity of diarrhea and need of toilet proximity were predictors of BAD in IBS-D (P

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Abbreviations used in this paper: BAD, bile acid diarrhea; BMI, body mass index; IBS, irritable bowel syndrome; IBS-D, irritable bowel syndrome with diarrhea.

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< .01). Patients with BAD were more likely to have a depression score higher than 8 on the Hospital Anxiety and Depression inventory.

CONCLUSIONS:

There is a greater impact on bowel and somatic symptoms and quality of life in IBS-D with BAD compared with IBS-D without BAD. Screening for BAD in IBS-D is especially relevant, with more severe and frequent diarrhea along with urgency.

Keywords: Malabsorption; Depression; Somatization.

Irritable bowel syndrome (IBS) is a functional gastrointestinal disorder affecting up to 10% of the general population.¹ The diagnosis is based on clinical criteria, because IBS is defined as recurrent abdominal pain related to defecation and change in the frequency or form of stool. IBS is subclassified into IBS with constipation, IBS with diarrhea (IBS-D), IBS with a mixed stool pattern, or IBS unclassified.² Up to 31.5% of people with IBS have IBS-D.³

IBS is associated with several predisposing factors such as antecedent infections, genetic predisposition, and psychological factors. Important gut factors include neuromuscular dysfunction, visceral hypersensitivity, increased permeability, dysbiosis, and bile acid-induced diarrhea (BAD), reflecting overall gut-brain axis dysfunction.^{4,5} IBS is associated with comorbidities in the form of other functional and somatic disorders such as fibromyalgia, chronic fatigue syndrome, and chronic pelvic pain, as well as psychiatric disorders such as major depressive disorder, anxiety, and somatization.⁶ Consequently, IBS is associated with decreased work productivity, decreased quality of life, and high health and financial burden.⁷

Although there is increased appreciation of the role of BAD in the development of chronic diarrhea, notably IBS-D, with evidence of 25% to 50% of patients in different series showing evidence of BAD,^{8–11} the impact of BAD on symptoms and quality of life in patients presenting with IBS-D is unknown. Bile acids are amphipathic molecules that mediate lipid digestion and absorption. After being synthesized in the liver, the primary bile acids (cholic acid and chenodeoxycholic acid) are conjugated to taurine and glycine and secreted in bile. Approximately 95% of these primary bile acids are reabsorbed via the apical Na⁺-dependent bile acid transporter in the terminal ileum and reach the liver via the portal circulation, thereby constituting the enterohepatic circulation. The remaining bile acids are deconjugated and dehydroxylated by colonic bacteria, forming predominantly deoxycholic acid and lithocholic acid, and these subsequently are excreted in the feces.

In idiopathic BAD unassociated with intestinal mucosal diseases or ileal resection, as occurs in patients with IBS-D, there is evidence of decreased synthesis of fibroblast growth factor-19, which is a hormone in the portal circulation that down-regulates the hepatocyte synthesis of bile acids.^{12–14} Hence, deficiency of fibroblast growth factor-19 is associated with increased bile acid synthesis, which can be assessed indirectly by measuring the level of

a bile acid synthesis intermediate, 7 α -hydroxy-4-cholesten-3-one. Increased synthesis leads to a higher proportion of bile acids reaching the colon, resulting in increased mucosal permeability, increased colonic motility, and water secretion contributing to BAD.¹⁵

Given that the impact of BAD on the symptoms and quality of life in patients presenting with IBS-D is unknown, our hypothesis was that patients with IBS-D with or without BAD have similar demographics, bowel and somatic symptoms, and quality of life. Our aim was to compare the demographics, bowel and somatic symptoms, and quality of life of patients with IBS-D with or without BAD.

Methods

Participants, Regulatory Approval, and Design

Data reported here were obtained from participants with Rome III-positive criteria¹⁶ for IBS-D who were recruited to the Mayo Clinic Institutional Review Board-approved protocol (Institutional Review Board #16-001445). Participants signed informed consent for use of their electronic medical records for research. The observational study used standard validated questionnaires and biochemical measurements. Details regarding patient selection and recruitment are included in the [Supplementary Methods](#).

Demographics

Demographic information regarding participant age, sex, body mass index (BMI), race, ethnicity, history of cholecystectomy, and number and types of medications taken was retrieved from the electronic medical records.

Questionnaires

The Hospital Anxiety and Depression Questionnaire. Participants answered 14 questions, 7 related to anxiety symptoms and 7 related to depression symptoms. Each question was scored from 0 to 3. The sum of the scores for the 7 anxiety-related questions and the 7 depression-related questions was computed. A cut-off score of 8 on each for the subscales was considered consistent with anxiety or depression.^{17,18}

Validated Bowel Disease Questionnaire. Participants answered 110 questions about their bowel symptoms

and the impact on their activities. The questionnaire also included 16 questions related to somatic symptoms. These somatic symptom questions were in the *how often* and *how bothersome* format and were scored on a scale from 0 (not a problem) to 4 (occurs daily or extremely bothersome when occurs).¹⁹

General Quality of Life: Symptom Checklist-90 Questionnaire. Participants answered 90 questions, scored from 0 to 4. These 90 questions explored 9 different dimensions (somatization, obsessive-compulsive, interpersonal sensitivity, depression, anxiety, hostility, phobic anxiety, paranoid ideation, and psychotic). A global severity index (mean score of the 90 questions), a positive symptom distress index (mean score for questions with a score >0), and a positive symptoms total (number of questions in which score was >0) was computed. The mean score for each of the dimensions also was computed.²⁰

Irritable Bowel Syndrome Specific Quality of Life questionnaire. Participants answered 34 questions about their quality of life, scored from 1 to 5. These questions belonged to 8 different domains (dysphoria, interference with activity, body image, health worry, food avoidance, social reaction, sexual, and relationship).²¹ Using a transformation formula, the scores were transformed into a scale of 0 to 100. This formula was used to compute the scaled scores of the 8 different domains.²²

Criteria for Identification of Bile Acid Diarrhea

Identification of BAD was based on established biochemical diagnostic criteria, that is, fasting (before 9:00 AM because of diurnal variation) serum 7 α -hydroxy-4-cholesten-3-one greater than 52.5 ng/mL.²³

Statistical Analysis

The associations between nominal category questionnaire responses as well as history of cholecystectomy and BAD status were assessed using a chi-square test. The associations for continuous baseline characteristics of age, BMI, and Symptom Checklist-90 subscales were assessed using a 2-sample *t* test. A nonparametric Wilcoxon rank sum test was used to examine the bowel functions and IBS quality-of-life category questions.

A multiple variable logistic regression model was used to examine 6 a priori identified questions for association with BAD status. These 6 questions pertaining to the past year for each participant were as follows: use of antidiarrheals, loose or watery stools, experienced an urgent need to open your bowels that made you rush to the toilet, worrying about losing control of bowel, number of bowel movements per week, and importance of being near a bathroom. A parsimonious model was fit using backward selection to identify the final model, retaining variables with a *P* value less than .05. The α level was set at .05 for statistical significance. SAS

What You Need to Know

Background

Bile acid diarrhea (BAD) is identified in approximately a third of patients with irritable bowel syndrome with diarrhea (IBS-D). This study investigated the impact of BAD by an analysis of bowel and somatic symptoms and quality of life of patients with IBS-D with or without BAD.

Findings

Patients with BAD had more severe bowel dysfunction (number of bowel movements, urgency), impact on IBS-specific quality of life (need of toilet proximity), and depression compared with IBS-D without BAD. Patients with BAD were more likely to receive antidiarrheal agents, bile acid binders, and anti-hydrogen ion secretion agents.

Implications for patient care

There is greater impact on bowel and somatic symptoms and quality of life in IBS-D with BAD compared with IBS-D without BAD. Screening for BAD in IBS-D is especially relevant with more severe and frequent diarrhea along with urgency.

version 9.4 (SAS Institute Inc., Cary, NC) was used in all analyses.

Results

Demographics

Our study included 219 patients with IBS-D; among these, 44 had BAD. Of our total population, 79% were females. Patients with IBS-D with BAD were significantly older and had a higher BMI than those without BAD. Patients with BAD had a significantly higher proportion of cholecystectomy compared with those without BAD. We found no significant differences for sex (Table 1), race, or ethnicity (not shown).

Medication Use

There was no significant difference in the median number of medications used between the 2 groups (Table 1). However, the use of antidiarrheal medications was significantly higher among BAD subjects compared with IBS-D without BAD. The use of fiber was comparable, with no statistically significant differences between groups (data not shown). There was a higher prevalence of using antacid secretory agents (H₂-receptor antagonists and proton pump inhibitors; *P* = .0321) and bile acid binders (*P* = .0043) in the BAD group compared with the group with IBS-D without BAD.

Table 1. Demographics of Patients With IBS-D With and Without BAD

	IBS-D BAD	IBS-D without BAD	P value
Patients, n	44	173	
Demographics			
Age, y, mean (SD)	46.3 (14.3)	39.8 (14.0)	.006
Sex, male/female, %	15.9/84.1	22.3/77.7	.353
BMI, kg/m ² , mean (SD)	33.3 (7.3)	29.2 (6.9)	<.001
Cholecystectomy	16 (36)	20 (11)	<.001
Gastrointestinal medications used			
Total patients, n	4.3 (2.3)	3.9 (2.3)	.604
Antidiarrheals	15 (34.1%)	31 (17.9%)	.019
Bile acid binders	5 (11.4%)	2 (1.2%)	.004
Laxative	1 (2.3%)	1 (0.6%)	.365
Antacid secretory agents	15 (34.1%)	33 (19.1%)	.032
Central neuromodulators used			
Antidepressants ^a	16 (36.4%)	68 (39.3%)	.721
TCA	1 (2.3%)	5 (2.9%)	1.0
SSRI	14 (31.8%)	53 (30.6%)	.880
SNRI	2 (4.5%)	7 (4.0%)	.880
Dopaminergic/noradrenergic reuptake inhibitors	4 (9.1%)	10 (5.8%)	.490
α ₂ adrenergic antagonists	0 (0.0%)	1 (0.6%)	1.0
Dopamine agonist	3 (6.8%)	1 (0.6%)	.027
Opioid agonist	1 (2.3%)	1 (0.6%)	.365
Other medications			
Eye lubricant	5 (11.4%)	3 (1.7%)	.010

NOTE. Data are shown as the median (interquartile range) or n (%).

BAD, bile acid diarrhea; BMI, body mass index; IBS-D, irritable bowel syndrome with diarrhea; SNRI, serotonin and norepinephrine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

^aPatients taking any kind of antidepressants were counted in this category. Patients taking more than 1 type of antidepressant were counted once.

No significant differences were found between both groups in the use of central neuromodulators. Moreover, no significant differences were found between both groups when the use of antidepressant was stratified by class of antidepressant. Dopamine agonist medication use in the BAD group was higher than in the group with IBS-D without BAD (3 in the BAD group and 1 in the IBS-D without BAD group; $P = .0270$); the indication for these dopamine agonists was restless legs syndrome. There was a significant difference in the use of eye lubricants between the BAD and non-BAD groups (5 in the BAD group and 4 in the IBS-D without BAD group).

Those medications that participants were asked to withhold from taking during the study are listed in [Supplementary Table 1](#).

Symptoms

Gastrointestinal and somatic symptoms. Patients with BAD were more likely to report more frequent, urgent, and watery bowel movements when compared with patients with IBS-D without BAD. For straining severity, patients with IBS-D with BAD reported a significantly lower prevalence of straining and seeing mucus in stool with bowel movements in the past year than those with IBS-D without BAD ([Table 2](#)).

No significant differences were found between the 2 groups in their experiences of somatic symptoms, and

frequency or intensity of bowel movements. Patients in the IBS-D with BAD group were more likely to smoke cigarettes, which was defined as at least 1 cigarette per day for at least 30 days, when compared with IBS-D without BAD.

Quality of Life

General quality of life based on Symptom Checklist-90. No significant differences were found in the global severity index, positive symptom distress index, and total positive symptoms between the 2 groups. Similarly, there were no significant differences between the mean scores on each of the 9 dimensions between the 2 groups ([Table 3](#)).

Irritable Bowel Syndrome Specific Quality of Life questionnaire. Significant differences between the 2 groups were found concerning the importance of being in proximity to a toilet, the worry about losing control of bowels (both were more prevalent in BAD group), and the fear of not being able to have a bowel movement (which was less likely to be present in the BAD group) ([Table 4](#)). Of the 8 subscales (see the Methods section), only interference with activity was significantly different between the 2 groups ($P = .0375$): BAD patients scored 39.3 (interquartile range, 23.3–60.7) compared with 33 (interquartile range, 17.9–50) for IBS-D without BAD, based on a scale of 0 to 100.

Table 2. Bowel Functions

	IBS-D with BAD	IBS-D without BAD	P value
Patients, n	44	160	
Bowel function			
Stools loose or watery in past year			
Sometimes	5 (12.2)	29 (19.5)	.002
Often	11 (26.8)	74 (49.7)	
Usually	25 (61.0)	46 (30.9%)	
Experienced an urgent need to open your bowels that made you rush to the toilet in the past year			
Sometimes	8 (19.5)	53 (35.6)	.002
Often	19 (46.3)	76 (51.0)	
Usually	14 (34.1)	18 (12.1)	
Mucus in stools in the past year	8 (25.0)	61 (46.9)	.029
Bowel movements/wk, n			
17–21	11 (26.8)	20 (13.4)	.001
22–26	6 (14.6)	19 (12.8)	
>26	9 (22.0)	16 (10.7)	
Severity of straining with BM			
Never	18 (43.9)	35 (23.6)	.049
Very mild	9 (22.0)	38 (25.7)	
Mild	4 (9.8)	29 (19.6)	
Moderate	7 (17.1)	40 (27.0)	
Severe	2 (4.9)	5 (3.4)	
Very severe	1 (2.4)	1 (0.7)	

NOTE. Data are shown as n (%).

BAD, bile acid diarrhea; BM, bowel movement; IBS-D, irritable bowel syndrome with diarrhea.

Hospital Anxiety and Depression scale. The questionnaire was completed by 206 patients. There were no significant differences between the 2 groups regarding total anxiety score, total depression score, and total (anxiety plus depression) score. When a cut-off value of 8 was used for the depression subscore, 9.1% of patients with BAD were positive compared with only 1.9% of patients without BAD ($P = .0188$).

Multiple Variable Logistic Model for Prediction of Bile Acid Diarrhea Status

A final parsimonious model examining use of anti-diarrheal medications, frequency of bowel movements, frequency of loose bowel movements, urgency, importance of being near a bathroom, and worrying about losing control of bowel identified 2 variables for prediction of bile acid diarrhea status: frequency of bowel movements (odds ratio, 1.35; 95% CI, 1.08–1.68) and worrying about losing control of bowel (odds ratio, 1.42; 95% CI, 1.08–1.868). The area under the receiver operating characteristic curve for this final model was 0.69 (95% CI, 0.59–0.78).

Discussion

In this study, we showed that patients with IBS-D with or without BAD differ in their BMI, medication use, symptoms, and quality-of-life parameters. Patients with

BAD were older, were more likely to have undergone cholecystectomy, and had a higher BMI compared with patients without BAD. The higher BMI has been reported previously in prior studies of BAD.^{24,25}

Patients with BAD were found to take significantly more antacids compared with patients without BAD, consistent with the high prevalence of gastroesophageal

Table 3. General Quality of Life Based on SCL-90 Dimensions

Dimension on SCL-90	IBS-D with BAD	IBS-D without BAD	P value
Patients, n	44	161	
Somatization	0.5 (0.5)	0.4 (0.4)	.185
Compulsive	0.5 (0.5)	0.4 (0.4)	.682
Sensitivity	0.4 (0.7)	0.3 (0.5)	.628
Depression	0.5 (0.5)	0.4 (0.5)	.778
Anxiety	0.3 (0.5)	0.2 (0.3)	.468
Hostility	0.3 (0.4)	0.3 (0.4)	.831
Phobic	0.2 (0.4)	0.1 (0.3)	.676
Paranoid	0.2 (0.5)	0.1 (0.3)	.246
Psychoticism	0.1 (0.2)	0.1 (0.2)	.980

NOTE. Data show means (SD).

BAD, bile acid diarrhea; IBS-D, irritable bowel syndrome with diarrhea; SCL-90, Symptom Checklist-90.

Table 4. IBS Quality of Life

	IBS-D BAD (N = 44)	IBS-D without BAD (N = 161)	P value
It is important to be near a toilet because of my bowel problems			.0105
Not at all	0 (0.0%)	12 (7.7%)	
Slightly	9 (20.9%)	44 (28.4%)	
Moderately	9 (20.9%)	38 (24.5%)	
Quite a bit	9 (20.9%)	31 (20.0%)	
Extremely	15 (34.9%)	30 (19.4%)	
I worry about losing control of my bowels			.0029
Not at all	8 (18.6%)	41 (26.5%)	
Slightly	10 (23.3%)	57 (36.8%)	
Moderately	5 (11.6%)	32 (20.6%)	
Quite a bit	11 (25.6%)	15 (9.7%)	
A great deal	9 (20.9%)	10 (6.5%)	
I fear that I will not be able to have a bowel movement			.0457
Missing	1	6	
Not at all	39 (90.7%)	120 (77.4%)	
Slightly	4 (9.3%)	24 (15.5%)	
Moderately	0 (0.0%)	7 (4.5%)	
Quite a bit	0 (0.0%)	1 (0.6%)	
A great deal	0 (0.0%)	2 (1.3%)	

NOTE. Data are shown as n (%).

BAD, bile acid diarrhea; IBS, irritable bowel syndrome; IBS-D, irritable bowel syndrome with diarrhea.

reflux disease among patients with IBS, and the fact that up to 44% of patients with IBS receive proton pump inhibitor therapy.^{26,27} It is conceivable that bacterial overgrowth in the small intestine associated with proton pump inhibitor use may increase deconjugation of bile acids and decrease their reabsorption^{28,29}; further studies are required to explore this hypothesis.

In our study, a greater proportion of the BAD group was receiving bile acid binders. Medical records review of the 5 patients with BAD taking bile acid binders showed that in 4 patients, the indication was a diagnosis of BAD before their enrollment in this study, and the fifth patient was treated for postcholecystectomy diarrhea.

It is worth noting that there was no difference in the number of patients in each group being treated with dietary fiber or with metformin, which may cause BAD by inhibiting ileal bile acid reabsorption.^{30,31} Patients on metformin were included only if the diarrhea preceded metformin initiation, the dose was stable, and they had not experienced aggravation of the diarrhea by the medication.

We found a smaller number of patients with BAD reported mucus in their stools compared with patients without BAD. This finding was not expected because previous animal studies showed that chenodeoxycholic and deoxycholic acid perfused into the colon increased mucus secretion.^{32–35}

The observation that patients with BAD took more antidiarrheal medications than patients without BAD is consistent with more frequent bowel movements per week in the BAD group. In fact, among patients with BAD,

63% had more than 17 bowel movements per week, 22% had more than 26 bowel movements per week, 61% reported loose or watery stools more than 75% of the time, and 80.4% of patients with BAD reported an urgent need to defecate more than 25% of the time. In fact, 34.1% had an urgency to defecate more than 75% of the time.

Conversely, patients with BAD strained less with bowel movements. These findings are consistent with previous studies that reported more severe diarrhea in patients with BAD, based on ⁷⁵SeHCAT retention less than 10%,²⁵ and on an online survey conducted by the Bile Acid Malabsorption Support UK Study, which showed that up to 80% reported severe diarrhea and 85% reported urgency.³⁶

Consistent with the significantly worse diarrhea, patients with BAD reported the importance of being near a toilet and the fear of losing control of their bowels, which precluded them from engaging in normal daily activities. Other studies have shown that patients with IBS-D had worse disease-specific quality of life and more interference with daily activities, especially those that involve getting out of the home and taking place in locations with no proximity to bathrooms when compared with other subtypes of IBS.³⁷ Treatment of BAD with bile acid sequestrants significantly improved work absenteeism.³⁶

BAD also impacts mental health, specifically showing a higher prevalence of symptoms suggestive of depression measured by the Hospital Anxiety and Depression Scale, but not when assessed by the Symptom Checklist-90. The divergent result might be owing to the small proportion of patients with symptoms consistent with depression even in the BAD group.

The strengths of our study include novel comparison of the demographics, symptoms, and quality of life of IBS-D patients with and without BAD, and the more than 200 patients with IBS-D with typical demographic features including female sex, middle age, and active symptoms at the time of study. Our study highlights the importance of screening for BAD among patients with IBS-D, especially in those with greater severity of symptoms such as number of bowel movements, urgency, looser stool consistency, and insecurity associated with needing proximity to a bathroom, which interferes with daily activities. Two features (frequency and worrying about losing control of bowel movements) also proved highly relevant on the multiple variable logistic model.

The limitations of our study are that the results are based on validated questionnaires, which still constitute a subjective measurement, and the absence of correction for multiple comparisons in the interrogations in the multiple questionnaires. However, in a prior study, we reported objective evidence consistent with more severe diarrhea in BAD (fecal bile acid excretion, >2337 $\mu\text{mol}/48\text{ h}$), documented by significantly faster colonic transit at 24 and 48 hours as well as higher fecal fat excretion compared with patients with IBS-D with normal fecal bile acid excretion.³⁸ Another limitation of our observational

study was the assessment of the population of interest at a single point in time and hence the need for replication of our findings.

We conclude that patients with BAD had a higher BMI, took more antidiarrheal agents, and had more severe and frequent diarrhea, as well as urgency. These symptoms interfere with daily activities, and the impact is confirmed on a rigorous regression model. The study adds to the body of evidence, including impact on health care utilization,³⁹ that searching for evidence of BAD is relevant in patients with IBS-D, especially in those with more severe and frequent diarrhea associated with urgency.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at <https://doi.org/10.1016/j.cgh.2021.11.035>.

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Supplementary Methods

Patient Selection

E-mail invitations and public advertisement were used to recruit patients living within 120 miles of Mayo Clinic in Rochester, Minnesota. The database included approximately 1200 patients with IBS. A 1-page evaluation of gastrointestinal symptoms was used to screen patients, and those with symptoms consistent with the Rome III criteria for IBS were enrolled in the study. Patients with an alternative diagnosis such as celiac disease, inflammatory bowel disease, or cancer were excluded from the study.¹

Timeline

During visit 1, the patients answered the Hospital Anxiety and Depression questionnaire. Participants were asked to withhold medications on the exclusion list ([Supplementary Table 1](#)) 7 days before visit 2. During visit 2, participants returned their completed Irritable Bowel Syndrome Specific Quality of Life questionnaire, Symptom Checklist-90 questionnaire, and validated bowel disease questionnaires.

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Supplementary Table 1. List of Medications Prohibited to Use at Least for 7 Days Before Visit 2 and Until Completion of Study

Prohibited medications	Examples
Any treatment specifically taken for IBS-D	Loperamide, cholestyramine, alosetron
Drugs with a known pharmacologic activity at 5-HT ₄ , 5-HT _{2b} , or 5-HT ₃ receptors	Tegaserod, ondansetron, tropisetron, granisetron, dolasetron, mirtazapine, cilansetron
All narcotics	Codeine, morphine, propoxyphene, tramadol, either alone or in combination
Anticholinergic agents	Dicyclomine, hyoscyamine, propantheline
Systemic antibiotics	Rifaximin and metronidazole
GI preparations	Antinausea agents (eg, trimethobenzamide, promethazine, prochlorperazine, dimenhydrinate, hydroxyzine) Osmotic laxative agents (eg, lactulose, sorbitol, or PEG solutions such as MiraLAX [Bayer Corp., Whippany, NJ] and GlycoLax [Kremers Urban Pharmaceuticals, Princeton, NJ]) Prokinetic agents (eg, cisapride, metoclopramide, prucalopride, domperidone)
Oral anticoagulants	
Antimuscarinics	
Peppermint oil	

5-HT_{2b}, 5-hydroxytryptamine receptor 2b; 5-HT₃, 5-hydroxytryptamine receptor 3; 5-HT₄, 5-hydroxytryptamine receptor 4; GI, gastrointestinal; IBS-D, irritable bowel syndrome with diarrhea; PEG, polyethylene glycol.